# SODIUM ALGINATE

Pregnancy: safe
Sodium content

**CHEMICAL NAME** Sodium salt of alginic acid (polysaccharide from brown seaweed) + sodium bicarbonate ± calcium carbonate
**CLASS** Alginate raft-forming antireflux agent
**TRADE NAMES** Gaviscon · Gaviscon Advance · Peptac · Acidex

## THERAPEUTIC USES / INDICATIONS

- GORD and heartburn, including in pregnancy.
- Regurgitation and laryngopharyngeal reflux.
- Infant reflux (Gaviscon Infant).

## ADMINISTRATION

- **PO** liquid or chewable tablets after meals and at bedtime.
- Do not drink immediately after (disrupts raft).
- Infant sachets mixed with feeds.

## DOSAGES

- **Adult:** 10–20 mL liquid or 2–4 tablets after meals and at bedtime (up to 4 times daily).
- **Children 6–12 yr:** 5–10 mL after meals.
- **Infants:** 1 sachet mixed with feed (max 6 per 24 h).

## PHARMACOKINETICS

- **Onset:** minutes. **Duration:** up to 4 h.
- Not absorbed; forms gel raft on stomach contents.
- Excreted in faeces.

## PHARMACODYNAMICS

- In acid, alginate forms a viscous gel raft floating on gastric contents ($CO_2$ from bicarbonate) → physical barrier to reflux.
- Bicarbonate/carbonate neutralise acid.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Bloating, flatulence.
- Constipation or diarrhoea.
- Nausea.

### ADVERSE EFFECTS (SERIOUS)

- Sodium overload (heart failure, hypertension, renal impairment) — standard Gaviscon ~6 mmol $Na^+$/10 mL.
- Infants: constipation, hypernatraemia, bezoar (overuse).

## CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** sodium-restricted diet, heart failure, renal impairment; phenylketonuria (some tablets).

### KEY INTERACTIONS

- May ↓ absorption of other drugs; separate by 2 h.

## NURSING MANAGEMENT

- Give after meals and at bedtime.
- Check sodium content for cardiac/renal patients.
- Infants: correct mixing; do not use with thickened feeds.

## PATIENT EDUCATION

- Take after meals and before bed; avoid lying down after meals.
- Raise head of bed; avoid large late meals.
- Safe in pregnancy.

**KEY POINTS**

- Forms a raft that blocks reflux.
- Safe in pregnancy.
- After meals and at bedtime.
- Watch sodium in HF/renal patients.

**SOURCES CHECKED** **[520]** Gaviscon Advance Aniseed flavour Oral Suspension (UK SmPC) **[521]** Gaviscon Original Aniseed Relief (UK SmPC)
*Full references p. 432–457*